Differential Equations Workbook

Modeling with differential equations

krista king MATH

DIRECTION FIELDS AND SOLUTION CURVES

■ 1. Sketch the direction field of the differential equation.

$$y' = 2y + x$$

■ 2. Sketch the direction field of the differential equation, and the solution curve passing through $(1,1)$.

$$y' = -3y + x$$

■ 3. Sketch the direction field of the differential equation, and the solution curve passing through $(0,0)$.

$$y' = y^2 + x^2 - 1$$

■ 4. Sketch the direction field for the differential equation, and the solution curve passing through $(0,1)$.

$$y' = 2y^2 - x^2 - 2$$

■ 5. Sketch the direction field of the differential equation.

$$y' = \sin y + x$$

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6. Sketch the direction field of the differential equation.

$$y' = xy$$

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INTERVALS OF VALIDITY

■ 1. Find the interval of validity for the solution to the differential equation, given $y(3) = -4$.

$$(x^2 - 25)y' - 7y - \ln(10 - x) = 0$$

■ 2. Find the interval of validity for the solution to the differential equation, given $y(0) = 1$.

$$(x + 2)y' + xye^{3x} - 5 = 0$$

■ 3. Find the interval of validity for the solution to the differential equation, given $y(0) = -1$, and solve the initial value problem to verify that the interval of validity is correct.

$$(x^2 - 1)y' + 2xy = x + 1$$

■ 4. Find the interval of validity for the solution to the differential equation, given $y(0) = y_0$, for some positive y_0.

$$y' = 3x^2y^{\frac{4}{3}}$$

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■ 5. Find the interval of validity for the solution to the differential equation.

$$\frac{dy}{dx} = y^2 e^x$$

$$y(0) = \frac{1}{e^2 - 1}$$

■ 6. Find the interval of validity for the solution to the differential equation, given $y(1) = y_0$.

$$xy' = y^2$$

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EULER'S METHOD

■ 1. If $y(0) = 0$, use Euler's method to approximate $y(2)$ with $n = 4$ steps.

$$y' = 1 + y$$

■ 2. If $y(2) = 0$, use Euler's method to approximate $y(4)$ with $n = 5$ steps.

$$y' = 3t + y$$

■ 3. If $y(0) = 1$, use Euler's method to approximate $y(1)$ with $n = 4$ steps.

$$y' = 1 - 2y$$

■ 4. Use Euler's method and five steps to approximate $y(3)$, given $y(2) = 1$.

$$y' = t^2 + y$$

■ 5. If $y(0) = 0$, use Euler's method to approximate $y(1)$ with $\Delta t = 0.2$.

$$y' + 3y = 1 - e^{-5t}$$

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6. If $y(0) = 3$, use Euler's method to approximate y_3 with $\Delta t = 0.1$, find the exact value and percentage error.

$$y' = 8e^{2t} - 3y$$

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AUTONOMOUS EQUATIONS AND EQUILIBRIUM SOLUTIONS

■ 1. Find any equilibrium solutions of the autonomous differential equation, then determine whether each solution is stable, unstable, or semi-stable without sketching a direction field.

$$\frac{dy}{dt} = -y^2 - y + 6$$

■ 2. Find any equilibrium solutions of the autonomous differential equation, then determine whether each solution is stable, unstable, or semi-stable without sketching a direction field.

$$\frac{dy}{dt} = y^3 - 4y^2 + 4y$$

■ 3. Find any equilibrium solutions of the autonomous differential equation, then determine whether each solution is stable, unstable, or semi-stable.

$$\frac{dy}{dt} = (y^2 - 9)(y - 1)^2$$

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■ 4. Find any equilibrium solutions of the autonomous differential equation, then determine whether each solution is stable, unstable, or semi-stable.

$$\frac{dy}{dt} = \sin y$$

■ 5. Find any equilibrium solutions of the autonomous differential equation, then determine whether each solution is stable, unstable, or semi-stable.

$$\frac{dy}{dt} = |16 - y^2|$$

■ 6. Find any equilibrium solutions of the autonomous differential equation, then determine whether each solution is stable, unstable, or semi-stable.

$$\frac{dy}{dt} + 3y + 6 = e^y(3y + 6)$$

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THE LOGISTIC EQUATION

■ 1. Describe the growth pattern of a population of butterflies, assuming their population is modeled by the logistic growth equation, if the carrying capacity of their habitat is $3{,}700$ butterflies.

■ 2. A bacteria population is observed to be $2{,}500$ at $t = 0$. Using the logistic growth model and a growth rate of 0.03 with a carrying capacity of $4{,}000$, write the logistic growth equation, then solve the initial value problem.

■ 3. An insect population is observed to be 500 at time $t = 0$. Assuming the population follows a logistic growth model with a growth rate of 0.01 and a carrying capacity of $2{,}500$, solve the initial value problem.

■ 4. The population of Canada in $2{,}000$ and 2010 was 30.7 and 34 million people, respectively. Predict its population in 2050 and 2060 assuming an exponential model of population growth.

■ 5. A bacteria population increases 8-fold in 5 hours. Assuming exponential growth, how long did it take for the population to increase 5-fold?

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■ 6. The growth rate of a population of bears is modeled by the differential equation, where time t is measured in years. Solve the differential equation with the initial condition $P(0) = 30$.

$$\frac{dP}{dt} = 0.002P(400 - P)$$

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PREDATOR-PREY SYSTEMS

■ 1. Is the system cooperative, competitive, or predator-prey? Find the equilibrium solutions of the system.

$$\frac{dx}{dt} = -0.16x + 0.08xy$$

$$\frac{dy}{dt} = 4.5y - 0.9xy$$

■ 2. Given the system modeling the interaction between populations of foxes and hares, find any equilibrium where the foxes and hares coexist, and give the size of each population.

$$\frac{dH}{dt} = 0.14H - 0.07FH$$

$$\frac{dF}{dt} = -0.65F + 0.05FH$$

■ 3. Determine whether the system is cooperative, competitive, or predator-prey, and find the maximum size of both populations when they are in equilibrium, assuming both populations are measured in millions.

$$\frac{dx}{dt} = 0.02x - 0.4x^2 + 0.005xy$$

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$$\frac{dy}{dt} = 0.1y - 1.1y^2 + 0.05xy$$

■ 4. Determine whether the system is cooperative, competitive, or predator-prey, and find the equilibrium solutions, assuming both populations are measured in millions.

$$\frac{dx}{dt} = 0.004x - 0.2xy$$

$$\frac{dy}{dt} = 0.03y - 0.36y^2 - 0.6xy$$

■ 5. Two species of bacteria are being cultivated in a petri dish. Every day, an amount of sugar is deposited in the petri dish and the species compete for this source of food. Determine whether the system is cooperative, competitive, or predator-prey and find the equilibrium solutions.

$$\frac{dx}{dt} = 0.001x - 0.03x^2 - 0.004xy$$

$$\frac{dy}{dt} = 0.002y - 0.05y^2 - 0.01xy$$

■ 6. Is the system cooperative, competitive, or predator-prey? Find the equilibrium solutions of the system.

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$$\frac{dx}{dt} = 0.48x - 0.16xy$$

$$\frac{dy}{dt} = 1.2y + 7.2xy$$

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EXPONENTIAL GROWTH AND DECAY

■ 1. If world population was 2.56 million in 1950 and 3.04 million in 1960 and is growing exponentially, find a function modeling population in the second half of the 20th century, then use the function to estimate world population in 1984.

■ 2. The size of a population of fleas increased by 4 fold in 2 months. How long did it take for the population to increase 5 fold?

■ 3. The half-life of radium-226 is $1{,}590$ years. If the substance decays exponentially, find a function that models the radioactive decay of a sample of 100 g of radium-226 over time t years. How much of the mass remains after $2{,}000$ years?

■ 4. A population of $1{,}000$ bacteria, growing exponentially, takes 12 minutes to double in size. How long will it take for the population to reach 1 million?

■ 5. A sofa's price drops by $50\,\%$ in 10 months. If the rate at which the price decreases is proportional to current price, how much will the price decrease in 2 years?

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■ 6. A radioactive element decays into nonradioactive substances. After 3 years, the radioactivity decreases by $5\,\%$. How many years would it take for the radioactivity to decrease by $90\,\%$?

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MIXING PROBLEMS

■ 1. A tank contains $5{,}000$ L of water and 100 kg of dissolved salt. Fresh water is entering the tank at 20 L/min, and the solution drains at a rate of 15 L/min. Assuming the solution in the tank remains perfectly mixed, what is the function that models the amount of salt $y(t)$ at any given minute t? How much salt remains in the tank after 15 days?

■ 2. A tank contains $2{,}000$ L of water and 500 kg of dissolved salt. Fresh water is entering the tank at 20 L/min, and the solution drains at a rate of 50 L/min. Assuming the solution in the tank remains perfectly mixed, what is the function $y(t)$ that models the amount of salt in the tank after t minutes?

■ 3. A tank contains $5{,}000$ L of freshwater and is being filled with a brine mixture with a concentration of 150 g/L, at a rate of 20 L/min. Meanwhile, solution drains from the tank at a rate of 30 L/min. Assuming the solution in the tank remains perfectly mixed, what is the function $y(t)$ that models the amount of salt in the tank after t minutes?

■ 4. A tank contains $1{,}000$ L of fresh water. A mix of water and salt is being added into the tank at a rate of 500 L/h, and the solution drains out at the same rate. The concentration of the solution that's being deposited varies

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periodically, according to the function $f(t) = 2 \sin t$ kg/L. Assuming that the solution in the tank remains perfectly mixed, how much salt is in the tank after t hours?

■ 5. A tank holding $2{,}000$ L of fresh water is being filled at 400 L/h with a brine solution with a concentration of $f(t) = 5 + \cos(t/5)$ kg/L. Assuming that the solution in the tank remains perfectly mixed and drains at a rate equal to the fill rate, how much salt is in the tank after t hours?

■ 6. A tank contains $3{,}000$ L of water and 600 kg of dissolved salt. Fresh water is entering the tank at 10 L/min, and the solution drains out at the same rate. Assuming the solution in the tank remains perfectly mixed, what is the function that models the amount of salt $y(t)$ at any given minute t?

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NEWTON'S LAW OF COOLING

■ 1. The temperature of a soup dropped from $100°C$ to $95°C$ in 5 minutes. How long would it take for the temperature to drop to $40°C$ if ambient temperature is $25°C$?

■ 2. The temperature of a liquid dropped from $350°C$ to $300°C$ in 30 minutes. What temperature would the liquid reach in 3 hours if the temperature of the environment is $30°C$?

■ 3. The temperature of ice cream in a freezer is $-20°$. After two minutes outside the freezer, where the temperature is $35°$, the temperature of the ice cream increased to $-10°$. What will be the temperature of the ice cream after 10 minutes?

■ 4. The temperature in a room is $25°$, while outside temperature is $-20°$. After 5 minutes with the door open, the room temperature dropped to $20°$. What will the temperature be in the room after 15 minutes, and when will the room temperature drop to $5°$?

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■ 5. A piece of metal is placed in a $20°C$ room, and after 5 minutes the temperature of the metal is $120°C$. After another 15 minutes the metal's temperature is $75°C$. What was the initial temperature of the metal?

■ 6. Room temperature can only be measured at temperatures less than or equal to $25°C$. We can't currently measure room temperature, which means the room must be warmer than $25°C$. Meanwhile, outside temperature is $-5°C$. After 15 minutes with the window open, room temperature has dropped to $20°C$, and to $17°C$ after another 5 minutes. What was the initial temperature of the room?

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ELECTRICAL SERIES CIRCUITS

■ 1. If a battery supplies a constant voltage of 9 V, has an inductance of 3 h and a resistance of $3\ \Omega$, and assuming $i(0) = 0$, find the current after 120 seconds.

■ 2. Given $L = 10$ h, $R = 70\ \Omega$, $C = 0.01$ f, $E(t) = 0$, $q(0) = 0$, and $i(0) = 3$, find the charge $q(t)$ on the capacitor.

■ 3. An inductance of 2 h is connected in series with a resistance of $10\ \Omega$ and a battery giving $E(t) = 120$ V. Initially, the current is zero. Formulate and solve an initial value problem modeling the electrical circuit.

■ 4. Suppose $L = 2$ h, $R = 6\ \Omega$, $E = 24e^{3t}$ V, and $i(0) = 0$. Formulate and solve an initial value problem that models the electrical circuit.

■ 5. Suppose $L = 2$ h, $R = 40\ \Omega$, $C = 0.005$ f, $E = 100$ V, $q(0) = 9$ C, and $q'(0) = i(0) = 0$. Formulate and solve an initial value problem that models this LRC circuit.

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6. An LRC circuit is set up with an inductance of $1/2$ h, a resistance of $1\ \Omega$, and a capacitance of $2/5$ f. Assuming the initial charge is 2 C and the initial current is 6 A, find the solution function describing the charge on the capacitor at any time t.

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SPRING AND MASS SYSTEMS

■ 1. Find the movement of a 6 kg mass attached to a spring with spring constant $k = 3$ kg/s^2 and damping constant $\beta = 6$ kg/s, given $x(0) = 2$ and $x'(0) = 0$.

■ 2. Assume an object weighing 8 lbs stretches a spring 12 in. Find the equation of motion if the spring is released from equilibrium with an upward velocity of $\sqrt{2}$ ft/sec.

■ 3. An 8 lb weight is attached to an 11 ft spring. When the mass comes to rest in the equilibrium position, the spring measures 15 ft. The system is immersed in a medium that imparts a damping force equal to $3/2$ times the instantaneous velocity of the mass. Find the equation of motion if the mass is pushed upward from the equilibrium position with an initial upward velocity of 4 ft/sec. What is the position of the mass after 15 sec?

■ 4. A 2 kg mass stretches a spring with a length of 20 cm. The system is attached to a dashpot that creates a damping force equal to 28 times the instantaneous velocity of the mass. Find the motion equation if the mass is released from rest at a point 6 cm below equilibrium.

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■ 5. A 16 lb weight stretches a spring 2 ft. Assume the damping force on the system is equal to the instantaneous velocity of the mass. Find the equation of motion of the mass.

■ 6. A mass of 9.8 kg stretches a spring 2.45 m. Find the equation that models the motion of the mass if we release the mass when $t = 0$ from a position 5 m above equilibrium, with a downward velocity of 2 m/s.

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